

Domain Discretization

To solve the governing diffusion equation numerically, the continuous spatial domain $0 \leq y \leq H$ is discretized into a finite number of grid points (nodes). Let N represent the total number of nodes along the y -direction. The uniform distance between consecutive nodes, referred to as the grid spacing Δy , is defined as:

$$\Delta y = \frac{H}{N - 1}$$

The discrete spatial coordinates of these grid points are given by:

$$y_j = j\Delta y, \quad j = 0, 1, 2, \dots, N - 1$$

In alignment with C++ indexing standards, the index j starts from 0 (bottom wall) and ends at $N - 1$ (top wall). The spatial discretization of the domain is illustrated in Figure 1.

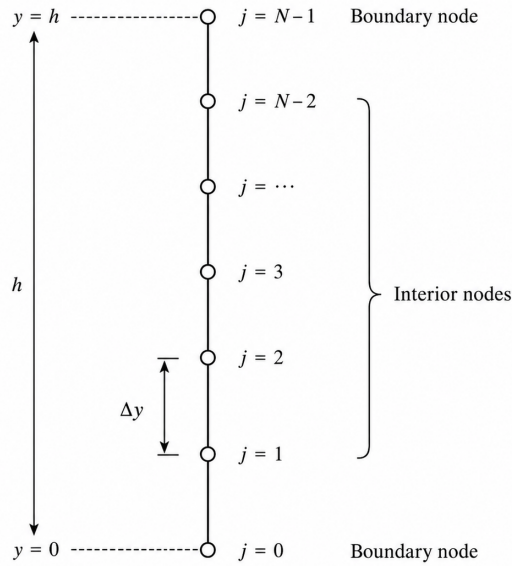


Figure 1: Computational Grid

Similarly, the temporal domain is discretized into uniform intervals of Δt , where n denotes the time level. The continuous velocity $u(y, t)$ at a specific grid location y_j and time $t^n = n\Delta t$ is represented using the following discrete notation:

$$u(y_j, t^n) \approx u_j^n$$

This discrete representation serves as the foundation for approximating the partial derivatives in the governing equation using Finite Difference Methods (FDM).